Micro

Step 1: Clone the original project

```
gugulothusrikanth@GugulothuSrikanth Desktop % git clone https://github.com/mohanDevOps-arch/Microservices-Task.git

Cloning into 'Microservices-Task'...
remote: Enumerating objects: 24, done.
remote: Counting objects: 100% (11/11), done.
remote: Compressing objects: 100% (8/8), done.
remote: Total 24 (delta 3), reused 3 (delta 3), pack-reused 13 (from 1)
Receiving objects: 100% (24/24), 12.42 KiB | 12.42 MiB/s, done.
Resolving deltas: 100% (3/3), done.
gugulothusrikanth@GugulothuSrikanth Desktop % cd Microservices-Task
```

Step 2: Check the project structure

```
gugulothusrikanth@GugulothuSrikanth Microservices-Task % ls -la
total 40
drwxr-xr-x   7 gugulothusrikanth  staff    224 Oct 17 22:07 .
drwx------@  7 gugulothusrikanth  staff    224 Oct 17 22:07 ..
drwxr-xr-x  12 gugulothusrikanth  staff    384 Oct 17 22:07 .git
-rw-r--r--   1 gugulothusrikanth  staff   3415 Oct 17 22:07 .gitignore
-rw-r--r--   1 gugulothusrikanth  staff  11357 Oct 17 22:07 LICENSE
drwxr-xr-x   6 gugulothusrikanth  staff    192 Oct 17 22:07 Microservices
-rw-r--r--   1 gugulothusrikanth  staff   1582 Oct 17 22:07 README.md
gugulothusrikanth@GugulothuSrikanth Microservices-Task % ls -la Microservices
total 0
drwxr-xr-x  6 gugulothusrikanth  staff  192 Oct 17 22:07 .
drwxr-xr-x  7 gugulothusrikanth  staff  224 Oct 17 22:07 ..
drwxr-xr-x  4 gugulothusrikanth  staff  128 Oct 17 22:07 gateway-service
drwxr-xr-x  4 gugulothusrikanth  staff  128 Oct 17 22:07 order-service
drwxr-xr-x  4 gugulothusrikanth  staff  128 Oct 17 22:07 product-service
drwxr-xr-x  4 gugulothusrikanth  staff  128 Oct 17 22:07 user-service
gugulothusrikanth@GugulothuSrikanth Microservices-Task % find Microservices -maxdepth 3 -type f | sort
Microservices/gateway-service/app.js
Microservices/gateway-service/package.json
Microservices/order-service/app.js
Microservices/order-service/package.json
Microservices/product-service/app.js
Microservices/product-service/package.json
Microservices/user-service/app.js
Microservices/user-service/package.json
gugulothusrikanth@GugulothuSrikanth Microservices-Task %
```

Step 3- Confirm the application ports and startup scripts

```
for service in user-service product-service order-service gateway-service; do
  echo ""
  echo "========== $service: package.json =========="
  cat "Microservices/$service/package.json"

  echo ""
  echo "========== $service: app.js =========="
  cat "Microservices/$service/app.js"
done

========== user-service: package.json ==========
{
    "name": "microservice",
    "version": "1.0.0",
    "dependencies": {
      "express": "^4.17.1",
      "axios": "^0.24.0"
    }
  }
========== user-service: app.js ==========
const express = require('express');
const app = express();
const port = 3000;

app.get('/health', (req, res) => {
  res.json({ status: 'User Service is healthy' });
});

app.get('/users', (req, res) => {
  const users = [
    { id: 1, name: 'John Doe' },
    { id: 2, name: 'Jane Smith' }
  ];
  res.json(users);
});

app.listen(port, () => {
  console.log(`User service running on port ${port}`);
});
========== product-service: package.json ==========
{
    "name": "microservice",
    "version": "1.0.0",
    "dependencies": {
      "express": "^4.17.1",
      "axios": "^0.24.0"
    }
  }
========== product-service: app.js ==========
const express = require('express');
const app = express();
const port = 3001;

app.get('/health', (req, res) => {
  res.json({ status: 'Product Service is healthy' });
});

app.get('/products', (req, res) => {
```

```
    "axios": "^0.24.0"
  }
}
========== order-service: app.js ==========
const express = require('express');
const axios = require('axios');
const app = express();
app.use(express.json());
const port = 3002;

const orders = [];

app.get('/health', (req, res) => {
  res.json({ status: 'Order Service is healthy' });
});

app.get('/orders', (req, res) => {
  res.json(orders);
});

app.post('/orders', async (req, res) => {
  const order = {
    id: orders.length + 1,
    userId: req.body.userId,
    productId: req.body.productId,
    timestamp: new Date()
  };
  orders.push(order);
  res.json(order);
});

app.listen(port, () => {
  console.log(`Order service running on port ${port}`);
});

========== gateway-service: package.json ==========
{
    "name": "microservice",
    "version": "1.0.0",
    "dependencies": {
      "express": "^4.17.1",
      "axios": "^0.24.0"
    }
  }
========== gateway-service: app.js ==========
const express = require('express');
const axios = require('axios');
const app = express();
app.use(express.json());
const port = 3003;

app.get('/health', (req, res) => {
  res.json({ status: 'Gateway Service is healthy' });
});

app.get('/api/users', async (req, res) => {
  try {
    const response = await axios.get('http://user-service:3000/users');
    res.json(response.data);
  } catch (error) {
    res.status(500).json({ error: 'Error fetching users' });
  }
```

Step 4 — Create the Dockerfiles

```
gugulothusrikanth@GugulothuSrikanth ~ % >....
    order-service) port=3002 ;;
    gateway-service) port=3003 ;;
  esac

  cat > "Microservices/$service/Dockerfile" <<EOF
FROM node:20-alpine

WORKDIR /app

COPY package*.json ./
RUN npm install --omit=dev

COPY app.js ./

EXPOSE $port

CMD ["node", "app.js"]
EOF
done

echo "Created Dockerfiles:"
find Microservices -name Dockerfile -print

Created Dockerfiles:
Microservices/product-service/Dockerfile
Microservices/gateway-service/Dockerfile
Microservices/order-service/Dockerfile
Microservices/user-service/Dockerfile
```

Step 5 — Create `docker-compose.yml`

```
gugulothusrikanth@GugulothuSrikanth ~ % >....
      - "3002:3002"
    networks:
      - microservices-network

  gateway-service:
    build:
      context: ./Microservices/gateway-service
    ports:
      - "3003:3003"
    depends_on:
      - user-service
      - product-service
      - order-service
    networks:
      - microservices-network

networks:
  microservices-network:
    driver: bridge
EOF

cat docker-compose.yml

services:
  user-service:
    build:
      context: ./Microservices/user-service
    ports:
      - "3000:3000"
    networks:
      - microservices-network

  product-service:
    build:
      context: ./Microservices/product-service
    ports:
      - "3001:3001"
    networks:
      - microservices-network

  order-service:
    build:
      context: ./Microservices/order-service
    ports:
      - "3002:3002"
    networks:
      - microservices-network

  gateway-service:
    build:
      context: ./Microservices/gateway-service
    ports:
      - "3003:3003"
    depends_on:
      - user-service
      - product-service
      - order-service
    networks:
      - microservices-network

networks:
```

```
  microservices-network:
    driver: bridge
```

```
 microservices-network:
    driver: bridge
EOF

cat docker-compose.yml
```

Step 6 — Docker application

```
gugulothusrikanth@GugulothuSrikanth Microservices-Task % docker --version
Docker version 29.5.3, build d1c06ef
gugulothusrikanth@GugulothuSrikanth Microservices-Task % docker compose version
Docker Compose version v5.1.4
gugulothusrikanth@GugulothuSrikanth Microservices-Task % docker info
Client:
 Version:    29.5.3
 Context:    desktop-linux
 Debug Mode: false
 Plugins:
  agent: Docker AI Agent Runner (Docker Inc.)
    Version:  v1.73.0
    Path:     /Users/gugulothusrikanth/.docker/cli-plugins/docker-agent
  ai: Docker AI Agent - Ask Gordon (Docker Inc.)
    Version:  v1.25.0
    Path:     /Users/gugulothusrikanth/.docker/cli-plugins/docker-ai
  buildx: Docker Buildx (Docker Inc.)
    Version:  v0.34.1-desktop.1
    Path:     /Users/gugulothusrikanth/.docker/cli-plugins/docker-buildx
  compose: Docker Compose (Docker Inc.)
    Version:  v5.1.4
    Path:     /Users/gugulothusrikanth/.docker/cli-plugins/docker-compose
  debug: Get a shell into any image or container (Docker Inc.)
    Version:  0.0.47
    Path:     /Users/gugulothusrikanth/.docker/cli-plugins/docker-debug
  desktop: Docker Desktop commands (Docker Inc.)
    Version:  v0.3.0
    Path:     /Users/gugulothusrikanth/.docker/cli-plugins/docker-desktop
  dhi: CLI for managing Docker Hardened Images (Docker Inc.)
    Version:  v0.0.4
    Path:     /Users/gugulothusrikanth/.docker/cli-plugins/docker-dhi
  extension: Manages Docker extensions (Docker Inc.)
    Version:  v0.2.31
    Path:     /Users/gugulothusrikanth/.docker/cli-plugins/docker-extension
  init: Creates Docker-related starter files for your project (Docker Inc.)
    Version:  v1.4.0
    Path:     /Users/gugulothusrikanth/.docker/cli-plugins/docker-init
  mcp: Docker MCP Plugin (Docker Inc.)
    Version:  v0.42.2
    Path:     /Users/gugulothusrikanth/.docker/cli-plugins/docker-mcp
  model: Docker Model Runner (Docker Inc.)
    Version:  v1.2.1
    Path:     /Users/gugulothusrikanth/.docker/cli-plugins/docker-model
  offload: Docker Offload (Docker Inc.)
    Version:  v0.6.3
    Path:     /Users/gugulothusrikanth/.docker/cli-plugins/docker-offload
  pass: Docker Pass Secrets Manager Plugin (beta) (Docker Inc.)
    Version:  v0.1.4
    Path:     /Users/gugulothusrikanth/.docker/cli-plugins/docker-pass
  sandbox: Docker Sandbox (Docker Inc.)
    Version:  v0.12.0
    Path:     /Users/gugulothusrikanth/.docker/cli-plugins/docker-sandbox
  scout: Docker Scout (Docker Inc.)
    Version:  v1.21.0
    Path:     /Users/gugulothusrikanth/.docker/cli-plugins/docker-scout

Server:
 Containers: 1
  Running: 1
  Paused: 0
  Stopped: 0
 Images: 19
```

```
gugulothusrikanth@GugulothuSrikanth Microservices-Task % docker compose up --build -d
[+] Building 74.7s (37/37) FINISHED
 => [internal] load local bake definitions                                                
```

```
gugulothusrikanth@GugulothuSrikanth Microservices-Task % curl http://localhost:3000/health

{"status":"User Service is healthy"}%
gugulothusrikanth@GugulothuSrikanth Microservices-Task % curl http://localhost:3001/health

{"status":"Product Service is healthy"}%
gugulothusrikanth@GugulothuSrikanth Microservices-Task % curl http://localhost:3002/health

{"status":"Order Service is healthy"}%
gugulothusrikanth@GugulothuSrikanth Microservices-Task % curl http://localhost:3003/health
gugulothusrikanth@GugulothuSrikanth Microservices-Task % curl http://localhost:3002/health

{"status":"Order Service is healthy"}%
gugulothusrikanth@GugulothuSrikanth Microservices-Task % curl http://localhost:3000/users

[{"id":1,"name":"John Doe"},{"id":2,"name":"Jane Smith"}]%
gugulothusrikanth@GugulothuSrikanth Microservices-Task % curl http://localhost:3001/products

[{"id":1,"name":"Laptop","price":999},{"id":2,"name":"Phone","price":699}]%
gugulothusrikanth@GugulothuSrikanth Microservices-Task % curl http://localhost:3002/orders

[]%
gugulothusrikanth@GugulothuSrikanth Microservices-Task % curl http://localhost:3002/orders

[]%
gugulothusrikanth@GugulothuSrikanth Microservices-Task % curl http://localhost:3003/api/users

[{"id":1,"name":"John Doe"},{"id":2,"name":"Jane Smith"}]%
gugulothusrikanth@GugulothuSrikanth Microservices-Task % curl http://localhost:3003/api/products

[{"id":1,"name":"Laptop","price":999},{"id":2,"name":"Phone","price":699}]%
gugulothusrikanth@GugulothuSrikanth Microservices-Task % curl http://localhost:3003/api/orders
[]%
gugulothusrikanth@GugulothuSrikanth Microservices-Task %
```

To test creating an order through the Gateway:

```
gugulothusrikanth@GugulothuSrikanth Microservices-Task % curl -X POST http://localhost:3003/api/orders \
  -H "Content-Type: application/json" \
  -d '{"userId":1,"productId":1}'
{"id":1,"userId":1,"productId":1,"timestamp":"2026-10-17T17:29:00.840Z"}%
gugulothusrikanth@GugulothuSrikanth Microservices-Task %



gugulothusrikanth@GugulothuSrikanth Microservices-Task % curl http://localhost:3003/api/orders
[{"id":1,"userId":1,"productId":1,"timestamp":"2026-10-17T17:29:00.840Z"}]%
gugulothusrikanth@GugulothuSrikanth Microservices-Task %
```

build command

```
 => CACHED [order-service 2/5] WORKDIR /app                                                
```

```
echo "=== Health Checks ==="
for port in 3000 3001 3002 3003; do
  echo "Port $port:"
  curl -sS "http://localhost:$port/health"
  echo
done

echo "=== Gateway Integration Tests ==="
curl -sS http://localhost:3003/api/users
echo
curl -sS http://localhost:3003/api/products
echo
curl -sS http://localhost:3003/api/orders
echo
=== Health Checks ===
Port 3000:
{"status":"User Service is healthy"}
Port 3001:
{"status":"Product Service is healthy"}
Port 3002:
{"status":"Order Service is healthy"}
Port 3003:
{"status":"Gateway Service is healthy"}
=== Gateway Integration Tests ===
[{"id":1,"name":"John Doe"},{"id":2,"name":"Jane Smith"}]
[{"id":1,"name":"Laptop","price":999},{"id":2,"name":"Phone","price":699}]
[]
gugulothusrikanth@GugulothuSrikanth Microservices-Task %
```